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Titans: Learning to Memorize at Test Time

Motivation. Transformers achieve strong performance due to attention-based context retrieval, but they are fundamentally limited by their finite context window and quadratic complexity with respect to sequence length. As a result, they struggle with very long contexts and cannot persistently store information beyond the active attention window.

The Titans architecture introduces a neural long-term memory module that allows models to *learn to memorize at test time*. The goal is to combine the strengths of attention (accurate short-term dependency modeling) with a persistent memory mechanism that can store information over extremely long time horizons.

Core Idea. The paper views sequence models through a memory lens:

- Attention acts as **short-term memory**.
- Recurrent memory modules act as **long-term memory**.

Titans integrate both components into a unified architecture where attention handles the local context and a separate memory module accumulates information over long time spans. This allows the model to scale to contexts far beyond traditional Transformer limits. [**?**]

Neural Long-Term Memory. The key contribution is a neural memory module that stores information dynamically during inference. Unlike standard neural networks whose parameters remain fixed during test time, the memory module updates itself using the incoming data stream.

The system measures the *surprise* of an input token using gradients of an associative memory objective. Tokens that violate model expectations are considered more informative and are preferentially stored in memory.

Memory updates resemble gradient-based learning rules:

$$M_t = M_{t-1} - \eta \nabla L(M_{t-1}; x_t)$$

where M_t is the memory state and L is a memory objective that captures key–value associations.

Memory Management. To prevent uncontrolled memory growth, the architecture introduces a decay mechanism similar to weight decay in optimization. This mechanism balances:

- the amount of stored information

- the surprise level of incoming data
- the total memory capacity

This effectively implements a learned forgetting mechanism.

Titan Architecture. The full model consists of three interacting components:

- **Core module:** a short-context attention mechanism that processes the current sequence window.
- **Long-term memory module:** a neural memory that accumulates information over long horizons.
- **Persistent memory:** task-specific parameters representing global knowledge about the domain.

The paper introduces three variants that integrate the memory module in different ways: as an additional context source, as a network layer, or as a gated branch.

Results. Experiments show that Titans:

- outperform modern recurrent sequence models
- match or exceed Transformers with equivalent context windows
- scale to contexts exceeding two million tokens

Performance improvements are observed in language modeling, commonsense reasoning, time-series forecasting, genomics tasks, and long-context retrieval benchmarks. [?]

Limitations. Despite promising results, several limitations remain:

- The architecture introduces additional complexity compared to standard Transformers.
- Memory updates may introduce instability during long sequences.
- The memory module relies on heuristic design choices such as surprise-based storage.

Further work is required to evaluate scaling behavior in large language models.

Takeaway. Titans propose a new hybrid architecture combining attention-based short-term reasoning with a learnable long-term memory that updates during inference. This approach aims to bridge the gap between Transformers' powerful local modeling and the persistent memory capabilities of recurrent systems.